



DC – AI – Routing – Switching – Security – Wireless – Automation

EVPN E-Line on PTX10k platforms

An HPE Juniper Networking Techpost

Ramdas Machat

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HPE



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1 EVPN E-Line on PTX10k platforms

Ramdas Machat - 12/12/2022

Let's test EVPN ELINE/VPWS on Express4-based platforms playing the role of PE. In this article, we will describe the various approaches, the configurations and the instance scaling.

1.1 Introduction

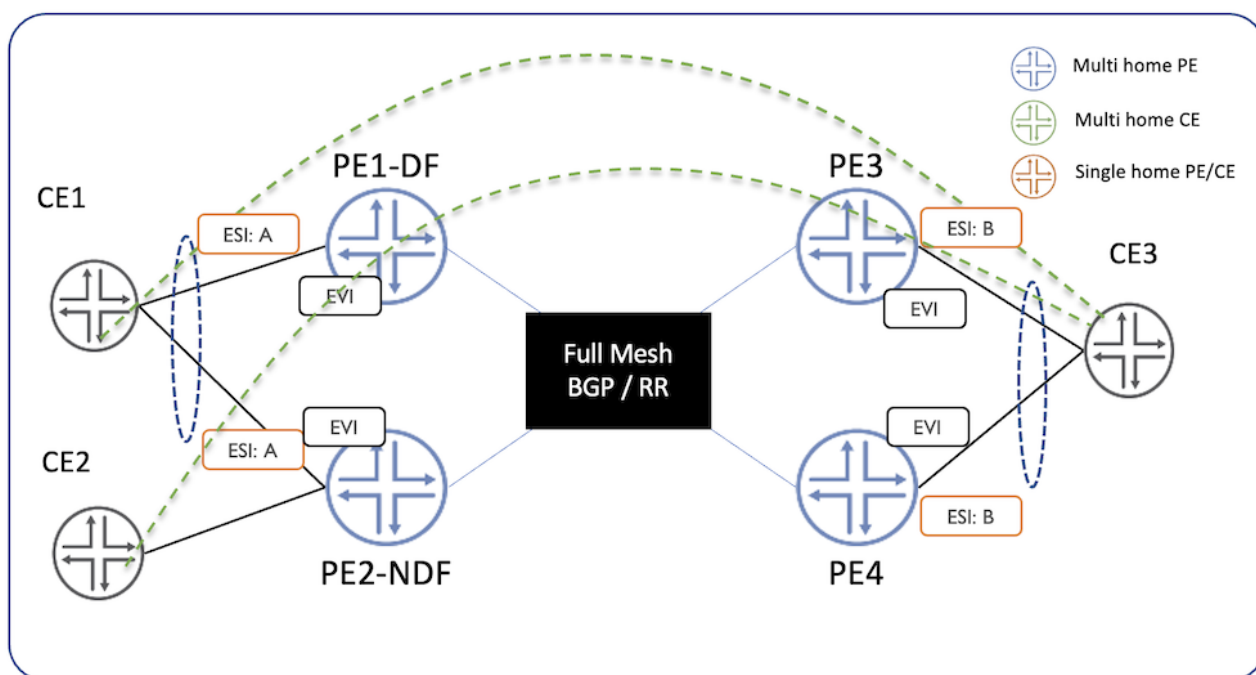
Service providers always look for the adoption of new technologies to improve their connectivity and the total uptime for the services they offer to their customers. There is a requirement to move to a more controlled and predictable network for the SP and Cloud providers. EVPN offers the most advanced L2 deployments with much better scale and performance in terms of availability, overcomes underutilization of revenue ports, faster convergence, and many more advantages over legacy technologies.

Ethernet EVPN (EVPN) as a technology has matured with large-scale deployment seen across various use case requirements including Service Provider and Cloud. EVPN helps with many benefits such as Faster convergence, Control plane MAC learning, Multiple Provider Edge (PE) participating in a single Ethernet Segment Identifier (ESI), and many more. Such features are not available in legacy L2 services.

1.2 Topology and Testing

This document covers the EVPN E-LINE service implementation on the PTX with Junos EVO. 400GE platforms powered by Express 4 ASIC were used to validate and produce the configurations.

Figure 1 is a reference topology and Figure 2 further down gives more details on the configuration and the different cases covered including Single homed, Multi-homed, and Scaling of the test scenario on a live topology with the configuration and operational status for each case.



Device Name	Model	Software Version
CE1	Hostname: R5-RE0Model: mx960	21.1R1.11
PE1	Hostname: r1-RE0-re0Model: ptx10008	22.2R2-202210200108.0-EVO
PE2	Hostname: r2-RE0-re0Model: ptx10008	22.2R2-202210200108.0-EVO
PE3	Hostname: r3-RE0-re0Model: ptx10008	22.2R2-202210200108.0-EVO
PE4	Hostname: r4-RE0-re0Model: ptx10008	22.2R2-202210200108.0-EVO
CE2	Hostname: R7-RE0Model: mx960	21.1R1.11

1.3 EVPN E-LINE/VPWS

EVPN-VPWS is a framework for delivering point-to-point VPN service with the EVPN signalling mechanisms. The endpoints are identified using a service identifier also called VPWS-ID discovered using BGP signaling. The service ids are 24bit values encoded in the Ethernet Tag of the Auto Discover (AD) route /Type 1 per EVI (Ethernet Virtual Instance). The advantages of EVPN VPWS are

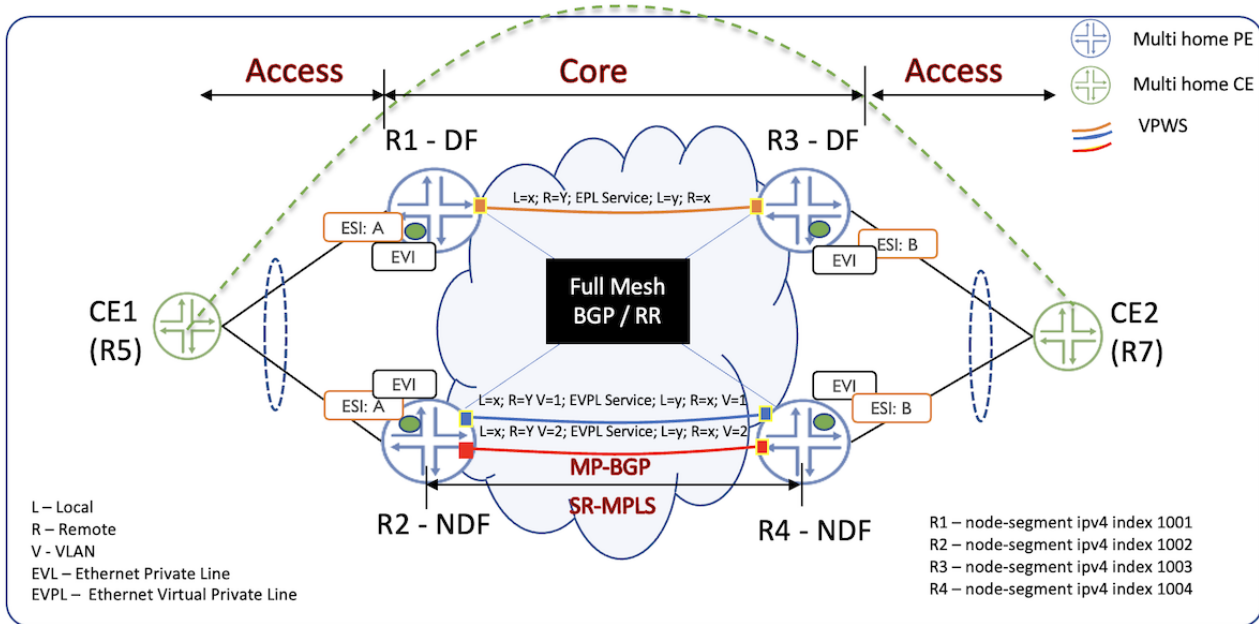
- Single homing (SH) and Multi homing (MH) scenario. One of the biggest advantages of the EVPN VPWS multi-homing is support for load balancing – both connections are active.
- Support for Inter-AS across the ASBR (Autonomous System Boundary Router) using the BGP EVPN signalled VPNs.

EVPN-VPWS is used to deliver private line services defined by the Metro Ethernet Forum (MEF)

- EPL (Ethernet Private Line)

Offers a point-to-point ethernet virtual connection (EVC) between a dedicated UNI (User Network Interface) with a high degree of transparency. - EVPL (Ethernet Virtual Private Line)

Offers service multiplexing which means multiple EVCs are paired per UNI



1.3.1 Underlay and Customer Equipment (CE) Configuration

All PEs as depicted in figure 2 are configured using the OSPF and MPLS for underlay or ISIS and SR. The PEs are configured with fully meshed BGP sessions with “family evpn signalling” enabled to exchange the EVPN routes among each other. A sample configuration is covered as part of Underlay Configuration – MPLS towards the end of the document for reference. Also, the CE configuration used during the testing is covered towards the end for reference.

1.4 EVPN VPWS EPL Test Scenarios

All four PEs in the above topology are BGP-EVPN PE and have the EVPN VPWS instance setup and then relevant configurations are made in each one of these PEs. These PEs discover each other in the network and established the EVPN sessions.

As covered above and as per the Metro Ethernet Forum standards, an EPL-based service is constructed based on the physical Ethernet Ports, and the service is provisioned between a pair of UNI ports.

The Ethernet Segment Identifier (ESI) is based on the Ethernet Virtual Instance (EVI) and port. In the Multi-homed scenario as depicted in the Topology above, the Designate Forwarder (DF) election here is Modulus (MOD) based (see [RFC7432](#)). As of time of writing this article, Express4 based PTX platforms support the preference-based DF election.

1.4.1 EVPN EPL Configuration – Single/Active mode

The below section highlights the EVPN Instance (EVI) and the UNI-facing interface configuration on all four PE nodes shown in figure 2 above for EVPN EVPL based service. This test focuses on the single-active mode.

PE1:

```

1 regress@r1-re0# show interfaces et-0/0/7
2 description "To R5";
3 encapsulation ethernet-ccc;
4 esi {
5     00:11:11:11:11:11:11:11:11:11;
6     single-active;
7 }
8 unit 0 {
9     family ccc;
10 }
11 regress@r1-re0# show routing-instances
12 evpn-vpws-epl {
13     instance-type evpn-vpws;
14     protocols {
15         evpn {
16             interface et-0/0/7.0 {
17                 vpws-service-id {
18                     local 1;
19                     remote 2;
20                 }
21             }
22         }
23     }
24     interface et-0/0/7.0;
25     route-distinguisher 11.11.11.11:1;
26     vrf-target target:11:1;
27 }

```

PE2:

```

1 regress@r2-re0# show interfaces et-0/0/9
2 description "To R5";
3 encapsulation ethernet-ccc;
4 esi {
5     00:11:11:11:11:11:11:11:11:11;
6     single-active;
7 }
8 unit 0 {
9     family ccc;
10 }
11 regress@r2-re0# show routing-instances
12 evpn-vpws-epl {
13     instance-type evpn-vpws;
14     protocols {
15         evpn {
16             interface et-0/0/9.0 {
17                 vpws-service-id {
18                     local 1;
19                     remote 2;
20                 }
21             }
22         }
23     }
24     interface et-0/0/9.0;
25     route-distinguisher 22.22.22.22:1;
26     vrf-target target:11:1;
27 }

```

PE3:

```

1 regress@r3-re0# show interfaces et-0/0/7
2 description "To R7";
3 encapsulation ethernet-ccc;
4 esi {
5     00:22:22:22:22:22:22:22:22:22;
6     single-active;
7 }
8 unit 0 {
9     family ccc;
10 }
11 regress@r3-re0# show routing-instances
12 evpn-vpws-epl {
13     instance-type evpn-vpws;
14     protocols {
15         evpn {
16             interface et-0/0/7.0 {
17                 vpws-service-id {
18                     local 2;

```

```

19         remote 1;
20     }
21 }
22 }
23 }
24 interface et-0/0/7.0;
25 route-distinguisher 33.33.33.33:1;
26 vrf-target target:11:1;
27 }

```

PE4:

```

1 regress@r3-re0# show interfaces et-0/0/7
2 description "To R7";
3 encapsulation ethernet-ccc;
4 esi {
5     00:22:22:22:22:22:22:22:22;
6     single-active;
7 }
8 unit 0 {
9     family ccc;
10 }
11 regress@r3-re0# show routing-instances
12 evpn-vpws-epl {
13     instance-type evpn-vpws;
14     protocols {
15         evpn {
16             interface et-0/0/7.0 {
17                 vpws-service-id {
18                     local 2;
19                     remote 1;
20                 }
21             }
22         }
23     }
24     interface et-0/0/7.0;
25     route-distinguisher 33.33.33.33:1;
26     vrf-target target:11:1;
27 }

```

1.4.2 EVPN EPL Configuration - Active/Active mode

The below section highlights the EVPN Instance (EVI) and the UNI-facing interface configuration on all four PE nodes shown in figure 2 above. This test focuses on the active-active mode.

In Active/Active we need to bundle the UNI links from CE to PE in order to load balance the traffic. So necessary configuration need to be done on the router to achieve the load balance behaviour.

Here we have a small change in the configurations to support active-active mode. The configuration shows just to add active-active on one of the MH PEs (PE1, PE2). But we cannot configure an MH PEs ESI in different modes like for example PE1 in active-standby and PE2 in active-active.

PE1:

```

1 regress@r1-re0# show interfaces ae0
2 ae0 {
3     encapsulation ethernet-ccc;
4     esi {
5         00:11:11:11:11:11:11:11:11;
6         single-active;
7     }
8     aggregated-ether-options {
9         lacp {
10             active;
11             system-id 00:00:00:00:11:01;
12         }
13     }
14     unit 0 {
15         family ccc;
16     }
17 }

```

```

18 regress@r1-re0# show routing-instances
19 evpn-vpws-evpl {
20     instance-type evpn-vpws;
21     protocols {
22         evpn {
23             interface ae0.0 {
24                 vpws-service-id {
25                     local 1;
26                     remote 2;
27                 }
28             }
29         }
30     }
31     interface ae0.0;
32     route-distinguisher 11.11.11.11:1;
33     vrf-target target:11:1;
34 }

```

PE2:

```

1 regress@r2-re0# show interfaces ae0
2 ae0 {
3     encapsulation ethernet-ccc;
4     esi {
5         00:11:11:11:11:11:11:11:11:11;
6         single-active;
7     }
8     aggregated-ether-options {
9         lacp {
10             active;
11             system-id 00:00:00:00:11:01;
12         }
13     }
14     unit 0 {
15         family ccc;
16     }
17 }
18 regress@r2-re0# show routing-instances
19 evpn-vpws-evpl {
20     instance-type evpn-vpws;
21     protocols {
22         evpn {
23             interface ae0.0 {
24                 vpws-service-id {
25                     local 1;
26                     remote 2;
27                 }
28             }
29         }
30     }
31     interface ae0.0;
32     route-distinguisher 22.22.22.22:1;
33     vrf-target target:11:1;
34 }

```

PE3:

```

1 regress@r3-re0# show interfaces ae0
2 ae0 {
3     encapsulation ethernet-ccc;
4     esi {
5         00:22:22:22:22:22:22:22:22:22;
6         single-active;
7     }
8     aggregated-ether-options {
9         lacp {
10             active;
11             system-id 00:00:00:00:22:01;
12         }
13     }
14     unit 0 {
15         family ccc;
16     }
17 }
18 regress@r3-re0# show routing-instances
19 evpn-vpws-evpl {
20     instance-type evpn-vpws;
21     protocols {
22         evpn {

```

```

23         interface ae0.1 {
24             vpls-service-id {
25                 local 2;
26                 remote 1;
27             }
28         }
29     }
30 }
31 interface ae0.1;
32 route-distinguisher 33.33.33.33:1;
33 vrf-target target:11:1;
34 }

```

PE4:

```

1 regress@r4-re0# show interfaces ae0
2 ae0 {
3     encapsulation ethernet-ccc;
4     esi {
5         00:22:22:22:22:22:22:22:22:22:22;
6         single-active;
7     }
8     aggregated-ether-options {
9         lacp {
10             active;
11             system-id 00:00:00:00:22:01;
12         }
13     }
14     unit 0 {
15         family ccc;
16     }
17 }
18 regress@r4-re0# show routing-instances
19 evpn-vpws-evpl {
20     instance-type evpn-vpws;
21     protocols {
22         evpn {
23             interface ae0.1 {
24                 vpws-service-id {
25                     local 2;
26                     remote 1;
27                 }
28             }
29         }
30     }
31     interface ae0.1;
32     route-distinguisher 44.44.44.44:1;
33     vrf-target target:11:1;
34 }

```

1.4.3 EVPN EPL Operational – Single/Active mode

The below cli outputs show information learned by the multihomed PEs for the ESI (Ethernet Segment Information) to which these PEs belong and then also the ESI information learned from the remote end. PE3, PE4 outputs are not captured here.

You can see that PE2 is DF (Designated Forwarder) and PE1 is Backup DF which were elected based on the MOD operation.

PE1:

```

1 regress@r1-re0# run show evpn vpw-Instance
2 Instance: evpn-vpws-epl, Instance type: EVPN VPWS
3 Route Distinguisher: 11.11.11.11:1
4 Number of local interfaces: 1 (1 up)
5
6 Interface name ESI Mode Role Status Control-Word Flow-Label-Tx Flow-Label-Rx
7 et-0/0/7.0 00:11:11:11:11:11:11:11:11:11single-active Backup Up Yes No No
8 Local SID: 1 Advertised Label: 104
9 PE addr ESI Label Mode Role TS Status
10 22.22.22.22 00:11:11:11:11:11:11:11:11 17 single-active Primary 2022-11-22
    22:18:59.989 Resolved

```

```

11 Remote SID: 2
12 PE addr ESI Label Mode Role TS Status
13 44.44.44.44 00:22:22:22:22:22:22:22 57 single-active Backup 2022-11-23
14 00:41:37.218 Resolved
15 33.33.33.33 00:22:22:22:22:22:22:22 91 single-active Primary 2022-11-23
16 00:41:36.999 Resolved
17 Number of protect interfaces: 0
18
19 Fast Convergence Information
20 ESI: 00:11:11:11:11:11:11:11 Number of PE nodes: 1
21 PE: 22.22.22.22
22 Advertised SID: 1
23
24 Fast Convergence Information
25 ESI: 00:22:22:22:22:22:22:22 Number of PE nodes: 2
26 PE: 44.44.44.44
27 Advertised SID: 2
28 PE: 33.33.33.33
29 Advertised SID: 2
30
31 DF Election Information for Single-Active ESI
32 ESI: 00:11:11:11:11:11:11:11
33 DF Election Algorithm: MOD based
34 SID used for DF Election: 1
35 Primary PE: 22.22.22.22
36 Backup PE: 11.11.11.11
37 Last DF Election: 2022-11-22 22:50:24
38
39 regress@r1-re0# run show interfaces et-0/0/7 terse
40 Interface Admin Link Proto Local Remote
41 et-0/0/7 up up
42 et-0/0/7.0 up down ccc
43
44 regress@r1-re0# run show route table mpls.0 ccc et-0/0/7
45 mpls.0: 23 destinations, 23 routes (23 active, 0 holddown, 0 hidden)
46 + = Active Route, - = Last Active, * = Both
47 et-0/0/7.0 *[EVPN/7] 02:39:42, route-type Egress
48 Discard
49
50 regress@r1-re0# run show route table mpls.0 ccc et-0/0/7 detail
51 mpls.0: 23 destinations, 23 routes (23 active, 0 holddown, 0 hidden)
52 et-0/0/7.0 (1 entry, 1 announced)
53 *EVPN Preference: 7
54 Next hop type: Discard, Next hop index: 0
55 Address: 0x564f543839dc
56 Next-hop reference count: 187, key opaque handle: (nil), non-key opaque handle: (nil)
57 )
58 State: <Active Int>
59 Age: 2:40:25
60 Validation State: unverified
61 Task: evpn global task
62 Announcement bits (1): 1-KRT
63 AS path: I
64 Route Type Egress
65 Thread: junos-main

```

PE2:

```

1 regress@r2-re0# run show evpn vpws-instance
2 Instance: evpn-vpws-epl, Instance type: EVPN VPWS
3 Route Distinguisher: 22.22.22.22:1
4 Number of local interfaces: 1 (1 up)
5
6 Interface name ESI Mode Role Status Control-Word Flow-Label-Tx Flow-Label-Rx
7 et-0/0/9.0 00:11:11:11:11:11:11:11 single-active Primary Up Yes No No
8 Local SID: 1 Advertised Label: 17
9 PE addr ESI Label Mode Role TS Status
10 11.11.11.11 00:11:11:11:11:11:11:11 104 single-active Backup 2022-11-22
11 22:19:00.204 Resolved
12 Remote SID: 2
13 PE addr ESI Label Mode Role TS Status
14 44.44.44.44 00:22:22:22:22:22:22:22 57 single-active Backup 2022-11-23
15 00:41:37.217 Resolved
16 33.33.33.33 00:22:22:22:22:22:22:22 91 single-active Primary 2022-11-23
17 00:41:36.998 Resolved
18 Number of protect interfaces: 0
19

```

```

17 Fast Convergence Information
18 ESI: 00:11:11:11:11:11:11:11:11 Number of PE nodes: 1
19 PE: 11.11.11.11
20 Advertised SID: 1
21
22 Fast Convergence Information
23 ESI: 00:22:22:22:22:22:22:22:22 Number of PE nodes: 2
24 PE: 44.44.44.44
25 Advertised SID: 2
26 PE: 33.33.33.33
27 Advertised SID: 2
28
29 DF Election Information for Single-Active ESI
30 ESI: 00:11:11:11:11:11:11:11:11
31 DF Election Algorithm: MOD based
32 SID used for DF Election: 1
33 Primary PE: 22.22.22.22
34 Backup PE: 11.11.11.11
35 Last DF Election: 2022-11-22 22:50:24
36
37 {master}[edit]
38
39 regress@r2-re0# run show interfaces et-0/0/9 terse
40 Interface Admin Link Proto Local Remote
41 et-0/0/9 up up
42 et-0/0/9.0 up up ccc
43
44 {master}[edit]
45 regress@r2-re0# run show route table mpls.0 ccc et-0/0/9 detail
46
47 mpls.0: 22 destinations, 22 routes (22 active, 0 holddown, 0 hidden)
48 et-0/0/9.0 (1 entry, 1 announced)
49 *EVPN Preference: 7
50 Next hop type: Indirect, Next hop index: 0
51 Address: 0x559b3cd59bdc
52 Next-hop reference count: 8, key opaque handle: (nil), non-key opaque handle: (nil)
53 Next hop type: Router, Next hop index: 23068
54 Next hop: 23.1.1.2 via et-0/0/1.0, selected
55 Label element ptr: 0x559b3d2a4868
56 Label parent element ptr: (nil)
57 Label element references: 3
58 Label element child references: 0
59 Label element lsp id: 0
60 Session Id: 5
61 Protocol next hop: 33.33.33.33
62 Label operation: Push 91 Offset: 252
63 Label TTL action: no-prop-ttl
64 Load balance label: Label 91: None;
65 Composite next hop: 0x559b3dea7100 23076 INH Session ID: 22
66 Indirect next hop: 0x559b390b7b08 23056 INH Session ID: 22
67 State: <Active Int>
68 Age: 22:39
69 Validation State: unverified
70 Task: evpn global task
71 Announcement bits (1): 1-KRT
72 AS path: I
73 Route Type Egress
74 Thread: junos-main

```

Traffic Screenshot showing traffic flowing.

Overview

Scenario

Ports

Chassis

Protocols

Protocol Interfaces

Static

Traffic

L2-3 Traffic Items

EPL Traffic

Overview

Scenario

Ports

Chassis

Protocols

Protocol Interfaces

Static

Traffic

L2-3 Traffic Items

EPL Traffic

Traffic

L2-3 Traffic Items

EPL Traffic

Flow Groups

Topology

Endpoint Sets

Transmit State

Suspend

Tx Port

Src/Dest Endpoint Pair

Rx Ports

Flow Group Name

Configured Frame Size

Applied Frame Size

Frame Rate

Tx port type

1

Traffic Item Name: EPL Traffic

Tx Mode: Interleaved, Src/Dst Mesh: OneToOne, Route Mesh: OneToOne, Bi-directional

CE2-R7

Src/Dest Endpoint P...

CE1-R5

EPL Traffic-EndpointSet-2 - Flow Group 0001

Fixed: 64

Fixed: 66

2000 f/s

Ethernet

2

Traffic Item Name: EPL Traffic

Tx Mode: Interleaved, Src/Dst Mesh: OneToOne, Route Mesh: OneToOne, Bi-directional

CE1-R5

Src/Dest Endpoint P...

CE2-R7

EPL Traffic-EndpointSet-2 - Flow Group 0002

Fixed: 64

Fixed: 66

2000 f/s

Ethernet

Summary

Flow groups

Frame Setup

Select Views...

Traffic Item Statistics

Port CPU Statistics

Port Statistics

Tx-Rx Frame Rate Statistics

Global Protocol Statistics

L2-L3 Test Summary Statistics

Traffic Item

Tx Frames

Rx Frames

Frames Delta

Loss %

Tx Frame Rate

Rx Frame Rate

Tx L1 Rate (bps)

Rx L1 Rate (bps)

Rx Bytes

Tx Rate (bps)

Rx Rate (bps)

Tx Rate (Kbps)

Rx Rate (Kbps)

Tx Rate (Mbps)

Rx Rate (Mbps)

EPL Traffic

85,835

85,801

34

0.040

4,000.392

4,061.391

2,752,269.936

2,794,236.807

5,662,866

264,025.895

268,051.787

2,112,207.361

2,144,414.294

2,112.207

2,144.414

2.112

2.144

1.4.4 EVPN EPL Operational - Active/Active mode

PE1:

```

1 Interface name ESI Mode Role Status Control-Word Flow-Label-Tx Flow-Label-Rx
2 ae0.1 00:11:11:11:11:11:11:11:11:11:11:11 all-active Primary Up Yes No No
3 Local SID: 1 Advertised Label: 131
4 PE addr ESI Label Mode Role TS Status
5 22.22.22.22 00:11:11:11:11:11:11:11:11:11 120 all-active Primary 2022-11-23
6 03:37:06.808 Resolved
7 Remote SID: 2
8 PE addr ESI Label Mode Role TS Status
9 44.44.44.44 00:22:22:22:22:22:22:22:22:22 57 all-active Primary 2022-11-23 03:37:08.809
10 Resolved
11 33.33.33.33 00:22:22:22:22:22:22:22:22:22 91 all-active Primary 2022-11-23 03:37:16.805
12 Resolved
13 Number of protect interfaces: 0
14 Fast Convergence Information
15 ESI: 00:11:11:11:11:11:11:11:11:11:11:11 Number of PE nodes: 1
16 PE: 22.22.22.22
17 Advertised SID: 1
18 Fast Convergence Information
19 ESI: 00:22:22:22:22:22:22:22:22:22:22:22 Number of PE nodes: 2
20 PE: 44.44.44.44
21 Advertised SID: 2
22 PE: 33.33.33.33
23 Advertised SID: 2
24 {master}[edit]
25 regress@r1-re0# run show interfaces ae0 terse
26 Interface Admin Link Proto Local Remote
27 ae0 up up
28 ae0.1 up up ccc
29 ae0.32767 up up multiservice
30
31 regress@r1-re0# run show route table mpls.0 ccc ae0.1 detail
32
33 mpls.0: 23 destinations, 23 routes (23 active, 0 holddown, 0 hidden)
34 ae0.1 (1 entry, 1 announced)
35 *EVPN Preference: 7
36 Next hop type: List, Next hop index: 25084
37 Address: 0x564f5011851c
38 Next-hop reference count: 3, key opaque handle: (nil), non-key opaque handle: (nil)
39 Next hop: ELNH Address 0x564f5395f99c
40 Next hop type: Indirect, Next hop index: 0
41 Address: 0x564f5395f99c
42 Next-hop reference count: 7, key opaque handle: (nil), non-key opaque handle: (
43 nil)
44 Protocol next hop: 33.33.33.33
45 Label operation: Push 91 Offset: 252
46 Composite next hop: 0x564f5395e840 25074 INH Session ID: 27
47 Indirect next hop: 0x564f4fc6b6988 25040 INH Session ID: 27
48 Next hop type: Router, Next hop index: 25076
49 Address: 0x564f54aab1c
50 Next-hop reference count: 7, key opaque handle: (nil), non-key opaque handle: (
51 nil)
52 Next hop: 13.1.1.2 via et-0/0/5.0
53 Label element ptr: 0x564f53ea43e8
54 Label parent element ptr: (nil)
55 Label element references: 3
56 Label element child references: 0
57 Label element lsp id: 0
58 Next hop: ELNH Address 0x564f53962b1c, selected
59 Next hop type: Indirect, Next hop index: 0
60 Address: 0x564f53962b1c
61 Next-hop reference count: 4, key opaque handle: (nil), non-key opaque handle: (
62 nil)
63 Protocol next hop: 44.44.44.44
64 Label operation: Push 57 Offset: 252
65 Composite next hop: 0x564f5395d880 25081 INH Session ID: 39
66 Indirect next hop: 0x564f57a24b08 25079 INH Session ID: 39
67 Next hop type: Router, Next hop index: 25080
68 Address: 0x564f54ab069c
69 Next-hop reference count: 7, key opaque handle: (nil), non-key opaque handle: (
70 nil)
71 Next hop: 14.1.1.2 via et-0/0/1.0
72 Label element ptr: 0x564f53ea43e8

```

```

69         Label parent element ptr: (nil)
70         Label element references: 3
71         Label element child references: 0
72         Label element lsp id: 0
73         State: <Active Int>
74         Age: 3:45
75         Validation State: unverified
76         Task: evpn global task
77         Announcement bits (1): 1-KRT
78         AS path: I
79         Route Type Egress
80         Thread: junos-main

```

1.5 EVPN VPWS EVPL Test Scenarios

All four PEs in the above topology are BGP-EVPN PE and have the EVPN VPWS instance setup and then relevant configurations are made in each one of these PEs. These PEs discover each other in the network and establishes the EVPN sessions.

We support both modes single/active and active-active offering EVPN VPWS service in multihomed PEs. The current configuration is with Single-Active for the Multihomed PEs.

As covered above and as per the Metro Ethernet Forum standards, an EVPL based service is constructed based on the IFD and the service is provisioned between a multiple pair of UNI ports. The ESI is based on the EVI and per IFD. In the Multihomed scenario as depicted in the Topology above, the DF election here is MOD based. Express4 based PTX platforms also support the preference-based DF election.

1.5.1 EVPN EVPL Configuration – Single/Active mode.

The below section highlights the EVPN Instance (EVI) and the UNI-facing interface configuration on all four PE nodes shown in figure 2 above for EVPN EVPL based service. This test focuses on the single-active mode.

PE1:

```

1 regress@r1-re0# show interfaces et-0/0/7
2 description "To R5";
3 flexible-vlan-tagging;
4 encapsulation vlan-ccc;
5 esi {
6     00:11:11:11:11:11:11:11:11:11;
7     single-active;
8 }
9 unit 1 {
10     encapsulation vlan-ccc;
11     vlan-id 1;
12 }
13
14 regress@r1-re0# show routing-instances evpn-vpws-epl
15 instance-type evpn-vpws;
16 protocols {
17     evpn {
18         interface et-0/0/7.1 {
19             vpws-service-id {
20                 local 1;
21                 remote 2;
22             }
23         }
24     }
25 }
26 interface et-0/0/7.1;
27 route-distinguisher 11.11.11.11:1;
28 vrf-target target:11:1;

```

PE2:

```

1 regress@r2-re0# show interfaces et-0/0/9
2 description "To R5";
3 flexible-vlan-tagging;
4 encapsulation vlan-ccc;
5 esi {
6     00:11:11:11:11:11:11:11:11:11;
7     single-active;
8 }
9 unit 1 {
10     encapsulation vlan-ccc;
11     vlan-id 1;
12 }
13
14 regress@r2-re0# show routing-instances evpn-vpws-epl
15 instance-type evpn-vpws;
16 protocols {
17     evpn {
18         interface et-0/0/9.1 {
19             vpws-service-id {
20                 local 1;
21                 remote 2;
22             }
23         }
24     }
25 }
26 interface et-0/0/9.1;
27 route-distinguisher 22.22.22.22:1;
28 vrf-target target:11:1;

```

1.5.2 EVPN EVPL Configuration – Active/Active mode

The below section highlights the EVPN Instance (EVI) and the UNI-facing interface configuration on all four PE nodes shown in figure 2 above for EVPN EVPL-based service. This test focuses on the active-active mode.

In Active/Active we need to bundle the UNI links from CE to PE in order to load balance the traffic. So necessary configuration needs to be done on the router to achieve the load balance behaviour.

Here we have a small change in the configurations to support active-active mode. The configuration shows just to add active-active on one of the MH PEs (PE1, PE2). But we cannot configure an MH PEs ESI in different modes, for example, PE1 in active-standby and PE2 in active-active.

PE1:

```

1 regress@r1-re0# show interfaces ae0
2 ae0 {
3     flexible-vlan-tagging;
4     encapsulation flexible-ethernet-services;
5     esi {
6         00:11:11:11:11:11:11:11:11:11;
7         all-active;
8     }
9     aggregated-ether-options {
10         lacp {
11             active;
12             system-id 00:00:00:00:11:01;
13         }
14     }
15     unit 1 {
16         encapsulation vlan-ccc;
17         vlan-id 1;
18     }
19 }
20
21 regress@r1-re0# show routing-instances
22 evpn-vpws-evpl {
23     instance-type evpn-vpws;
24     protocols {
25         evpn {
26             interface ae0.1 {
27                 vpws-service-id {
28                     local 1;

```

```

29         remote 2;
30     }
31 }
32 }
33 }
34 interface ae0.1;
35 route-distinguisher 11.11.11.11:1;
36 vrf-target target:11:1;
37 }

```

PE2:

```

1 regress@r2-re0# show interfaces ae0
2 ae0 {
3     flexible-vlan-tagging;
4     encapsulation flexible-ethernet-services;
5     esi {
6         00:11:11:11:11:11:11:11:11:11;
7         all-active;
8     }
9     aggregated-ether-options {
10         lacp {
11             active;
12             system-id 00:00:00:00:11:01;
13         }
14     }
15     unit 1 {
16         encapsulation vlan-ccc;
17         vlan-id 1;
18     }
19 }
20
21 regress@r2-re0# show routing-instances
22 evpn-vpws-evpl {
23     instance-type evpn-vpws;
24     protocols {
25         evpn {
26             interface ae0.1 {
27                 vpws-service-id {
28                     local 1;
29                     remote 2;
30                 }
31             }
32         }
33     }
34     interface ae0.1;
35     route-distinguisher 22.22.22.22:1;
36     vrf-target target:11:1;
37 }

```

1.5.3 EVPN EVPL Operational – Single/Active mode

PE1

```

1 regress@r1-re0# run show evpn vpwsw-instance
2 Instance: evpn-vpws-epl, Instance type: EVPN VPWS
3 Route Distinguisher: 11.11.11.11:1
4 Number of local interfaces: 1 (1 up)
5
6 Interface name ESI Mode Role Status Control-Word Flow-Label-Tx Flow-Label-Rx
7 et=0/0/7.1 00::11:11:11:11:11:11:11:11:11:11:11:single-active Backup Up Yes No No
8 Local SID: 1 Advertised Label: 104
9 PE addr ESI Label Mode Role TS Status
10 22.22.22.22 00::11:11:11:11:11:11:11:11:11:11:17 single-active Primary 2022-11-23
    01:34:14.940 Resolved
11 Remote SID: 2
12 PE addr ESI Label Mode Role TS Status
13 33.33.33.33 00::22:22:22:22:22:22:22:22:22:22:91 single-active Primary 2022-11-23
    01:51:10.610 Resolved
14 44.44.44.44 00::22:22:22:22:22:22:22:22:22:22:57 single-active Backup 2022-11-23
    01:52:40.485 Resolved
15 Number of protect interfaces: 0
16
17 Fast Convergence Information
18 ESI: 00::11:11:11:11:11:11:11:11:11:11: Number of PE nodes: 1

```

```

19      PE: 22.22.22.22
20      Advertised SID: 1
21
22      Fast Convergence Information
23      ESI: 00:22:22:22:22:22:22:22:22:22 Number of PE nodes: 2
24      PE: 33.33.33.33
25      Advertised SID: 2
26      PE: 44.44.44.44
27      Advertised SID: 2
28
29      DF Election Information for Single-Active ESI
30      ESI: 00:11:11:11:11:11:11:11:11:11
31      DF Election Algorithm: MOD based
32      SID used for DF Election: 1
33      Primary PE: 22.22.22.22
34      Backup PE: 11.11.11.11
35      Last DF Election: 2022-11-23 01:50:02
36
37      {master}[edit]
38      regress@r1-re0# run show interfaces et-0/0/7 terse
39      Interface Admin Link Proto Local Remote
40      et-0/0/7 up up
41      et-0/0/7.1 up down ccc
42      et-0/0/7.32767 up up multiservice
43
44      {master}[edit]
45      regress@r1-re0# run show route table mpls.0 ccc et-0/0/7.1
46
47      mpls.0: 23 destinations, 23 routes (23 active, 0 holddown, 0 hidden)
48      + = Active Route, - = Last Active, * = Both
49
50      et-0/0/7.1 *[EVPN/7] 01:00:57, route-type Egress
51      Discard
52
53      {master}[edit]
54      regress@r1-re0# run show route table mpls.0 ccc et-0/0/7.1 detail
55
56      mpls.0: 23 destinations, 23 routes (23 active, 0 holddown, 0 hidden)
57      et-0/0/7.1 (1 entry, 1 announced)
58      *EVPN Preference: 7
59      Next hop type: Discard, Next hop index: 0
60      Address: 0x564f543839dc
61      Next-hop reference count: 201, key opaque handle: (nil), non-key opaque handle: (nil)
62      State: <Active Int>
63      Age: 1:01:00
64      Validation State: unverified
65      Task: evpn global task
66      Announcement bits (1): 1-KRT
67      AS path: I
68      Route Type Egress
69      Thread: junos-main

```

PE2

```

1 regress@r2-re0# run show evpn vpws-instance
2 Instance: evpn-vpws-epl, Instance type: EVPN VPWS
3 Route Distinguisher: 22.22.22.22:1
4 Number of local interfaces: 1 (1 up)
5
6 Interface name ESI Mode Role Status Control-Word Flow-Label-Tx Flow-Label-Rx
7 et-0/0/9.1 00:11:11:11:11:11:11:11:11:11 single-active Primary Up Yes No No
8 Local SID: 1 Advertised Label: 17
9 PE addr ESI Label Mode Role TS Status
10 11.11.11.11 00:11:11:11:11:11:11:11:11:11 104 single-active Backup 2022-11-23
    01:50:02.447 Resolved
11 Remote SID: 2
12 PE addr ESI Label Mode Role TS Status
13 33.33.33.33 00:22:22:22:22:22:22:22:22:22 91 single-active Primary 2022-11-23
    01:51:10.610 Resolved
14 44.44.44.44 00:22:22:22:22:22:22:22:22:22 57 single-active Backup 2022-11-23
    01:52:40.488 Resolved
15 Number of protect interfaces: 0
16
17 Fast Convergence Information
18 ESI: 00:11:11:11:11:11:11:11:11:11 Number of PE nodes: 1
19 PE: 11.11.11.11
20 Advertised SID: 1
21
22 Fast Convergence Information

```

```

23 ESI: 00:22:22:22:22:22:22:22:22 Number of PE nodes: 2
24 PE: 33.33.33.33
25 Advertised SID: 2
26 PE: 44.44.44.44
27 Advertised SID: 2
28
29 DF Election Information for Single-Active ESI
30 ESI: 00:11:11:11:11:11:11:11
31 DF Election Algorithm: MOD based
32 SID used for DF Election: 1
33 Primary PE: 22.22.22.22
34 Backup PE: 11.11.11.11
35 Last DF Election: 2022-11-23 01:50:01
36
37 {master}[edit]
38 regress@r2-re0# run show interfaces et-0/0/9 terse
39 Interface Admin Link Proto Local Remote
40 et-0/0/9 up up
41 et-0/0/9.1 up up ccc
42 et-0/0/9.32767 up up multiservice
43
44 {master}[edit]
45 regress@r2-re0# run show route table mpls.0 ccc et-0/0/9.1 detail
46
47 mpls.0: 23 destinations, 23 routes (23 active, 0 holddown, 0 hidden)
48 et-0/0/9.1 (1 entry, 1 announced)
49 *EVPN Preference: 7
50 Next hop type: Indirect, Next hop index: 0
51 Address: 0x559b3cd59bdc
52 Next-hop reference count: 16, key opaque handle: (nil), non-key opaque handle: (nil)
53 Next hop type: Router, Next hop index: 23068
54 Next hop: 23.1.1.2 via et-0/0/1.0, selected
55 Label element ptr: 0x559b3d2a4868
56 Label parent element ptr: (nil)
57 Label element references: 3
58 Label element child references: 0
59 Label element lsp id: 0
60 Session Id: 5
61 Protocol next hop: 33.33.33.33
62 Label operation: Push 91 Offset: 252
63 Label TTL action: no-prop-ttl
64 Load balance label: Label 91: None;
65 Composite next hop: 0x559b3dea7100 23076 INH Session ID: 22
66 Indirect next hop: 0x559b390b7b08 23056 INH Session ID: 22
67 State: <Active Int>
68 Age: 59:41
69 Validation State: unverified
70 Task: evpn global task
71 Announcement bits (1): 1-KRT
72 AS path: I
73 Route Type Egress
74 Thread: junos-main

```

Traffic Screenshot

The screenshot displays the Traffic Tools interface for configuring and monitoring traffic. The main window shows the configuration for 'EVPL-Vlan-1' under 'L2-3 Traffic Items'. The configuration table lists two flow groups with their respective endpoints and rates.

Flow Group	Topology	Endpoint Sets	TX Mode	TX Rate	Rx Rate	Flow Group Name	Configured Frame Size	Applied Frame Size	Frame Rate	Tx port type
1	CE1-R1	CE1-R1	Interleaved, Src/Dst Mesh: OneToOne, Route Mesh: OneToOne, Bi-directional	2000 f/s	2000 f/s	EVPL-Vlan-1-EndpointSet-1-Flow Group 0001	Fixed: 64	Fixed: 70	2000 f/s	Ethernet
2	CE1-R1	CE1-R1	Interleaved, Src/Dst Mesh: OneToOne, Route Mesh: OneToOne, Bi-directional	2000 f/s	2000 f/s	EVPL-Vlan-1-EndpointSet-1-Flow Group 0002	Fixed: 64	Fixed: 70	2000 f/s	Ethernet

Below the configuration table, the 'Traffic Item Statistics' section provides a detailed view of the traffic flow. The table below shows the statistics for 'EVPL-Vlan-1'.

Traffic Item	Tx Frames	Rx Frames	Frames Delta	Loss %	Tx Frame Rate	Rx Frame Rate	Tx L1 Rate (Bps)	Rx L1 Rate (Bps)	Rx Bytes	Tx Rate (Bps)	Rx Rate (Bps)	Tx Rate (Kbps)	Rx Rate (Kbps)	Tx Rate (Mbps)	Rx Rate (Mbps)	Store-Forward Avg Latency (ms)
EVPL-Vlan-1	270,960	270,948	12	0.004	3,999.994	3,998.994	2,879,995.795	2,879,275.706	18,966,360	279,999.591	279,929.583	2,239,996.000	2,239,436.000	2,239.997	2,239.437	2.240

1.5.4 EVPN EVPL Operational – Active/Active mode

PE1

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```

74                                     Label parent element ptr: (nil)
75                                     Label element references: 3
76                                     Label element child references: 0
77                                     Label element lsp id: 0
78                               State: <Active Int>
79                               Age: 3:45
80                               Validation State: unverified
81                               Task: evpn global task
82                               Announcement bits (1): 1-KRT
83                               AS path: I
84                               Route Type Egress
85                               Thread: junos-main

```

PE2

```

1 regress@r2-re0# run show evpn vpws-instance
2 Instance: evpn-vpws-epl, Instance type: EVPN VPWS
3 Route Distinguisher: 22.22.22.22:1
4 Number of local interfaces: 1 (1 up)
5
6 Interface name ESI Mode Role Status Control-Word Flow-Label-Tx Flow-Label-Rx
7 ae0.1 00:11:11:11:11:11:11:11:11:11:11:11:11:all-active Primary Up Yes No No
8 Local SID: 1 Advertised Label: 120
9 PE addr ESI Label Mode Role TS Status
10 11.11.11.11 00:11:11:11:11:11:11:11:11:11 131 all-active Primary 2022-11-23
    03:37:07.242 Resolved
11 Remote SID: 2
12 PE addr ESI Label Mode Role TS Status
13 44.44.44.44 00:22:22:22:22:22:22:22:22:22 57 all-active Primary 2022-11-23 03:37:08.809
    Resolved
14 33.33.33.33 00:22:22:22:22:22:22:22:22:22 91 all-active Primary 2022-11-23 03:37:16.821
    Resolved
15 Number of protect interfaces: 0
16
17 Fast Convergence Information
18 ESI: 00:11:11:11:11:11:11:11:11:11:11:11 Number of PE nodes: 1
19 PE: 11.11.11.11
20 Advertised SID: 1
21
22 Fast Convergence Information
23 ESI: 00:22:22:22:22:22:22:22:22:22:22:22 Number of PE nodes: 2
24 PE: 44.44.44.44
25 Advertised SID: 2
26 PE: 33.33.33.33
27 Advertised SID: 2
28
29 {master}[edit]
30 regress@r2-re0# run show interfaces ae0 terse
31 Interface Admin Link Proto Local Remote
32 ae0 up up
33 ae0.1 up up ccc
34 ae0.32767 up up multiservice
35
36 {master}[edit]
37 regress@r2-re0# run show route table mpls.0 ccc ae0.1 detail
38
39 mpls.0: 23 destinations, 23 routes (23 active, 0 holddown, 0 hidden)
40 ae0.1 (1 entry, 1 announced)
41 *EVPN Preference: 7
42 Next hop type: List, Next hop index: 23085
43 Address: 0x559b3cdb71c
44 Next-hop reference count: 3, key opaque handle: (nil), non-key opaque handle: (nil)
45 Next hop: ELNH Address 0x559b3cd59bdc, selected
46 Next hop type: Indirect, Next hop index: 0
47 Address: 0x559b3cd59bdc
48 Next-hop reference count: 17, key opaque handle: (nil), non-key opaque handle: (
    nil)
49 Protocol next hop: 33.33.33.33
50 Label operation: Push 91 Offset: 252
51 Composite next hop: 0x559b3dea7100 23076 INH Session ID: 22
52 Indirect next hop: 0x559b390b7b08 23056 INH Session ID: 22
53 Next hop type: Router, Next hop index: 23068
54 Address: 0x559b3dea9b1c
55 Next-hop reference count: 7, key opaque handle: (nil), non-key opaque handle:
    (nil)
56 Next hop: 23.1.1.2 via et-0/0/1.0
57 Label element ptr: 0x559b3d2a4868
58 Label parent element ptr: (nil)
59 Label element references: 3
60 Label element child references: 0

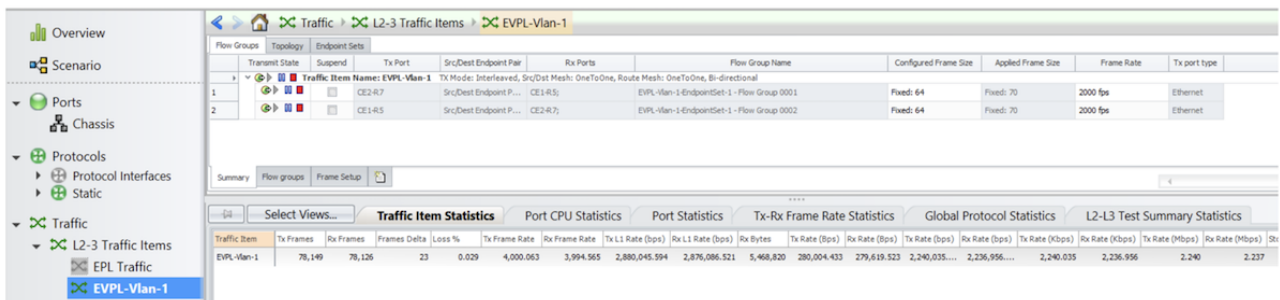
```

```

61      Label element lsp id: 0
62      Next hop: ELNH Address 0x559b3cd5bf1c
63      Next hop type: Indirect, Next hop index: 0
64      Address: 0x559b3cd5bf1c
65      Next-hop reference count: 31, key opaque handle: (nil), non-key opaque handle: (
        nil)
66      Protocol next hop: 44.44.44.44
67      Label operation: Push 57 Offset: 252
68      Composite next hop: 0x559b3cd586c0 23089 INH Session ID: 15
69      Indirect next hop: 0x559b390b7d88 23061 INH Session ID: 15
70      Next hop type: Router, Next hop index: 23090
71      Address: 0x559b3deac5dc
72      Next-hop reference count: 7, key opaque handle: (nil), non-key opaque handle:
        (nil)
73      Next hop: 24.1.1.2 via et-0/0/5.0
74      Label element ptr: 0x559b3d2a4868
75      Label parent element ptr: (nil)
76      Label element references: 3
77      Label element child references: 0
78      Label element lsp id: 0
79      State: <Active Int>
80      Age: 6:01
81      Validation State: unverified
82      Task: evpn global task
83      Announcement bits (1): 1-KRT
84      AS path: I
85      Route Type Egress
86      Thread: junos-main

```

Traffic Screenshot



Transmit State	Suspend	Tx Port	Src/Dest Endpoint Pair	Rx Ports	Flow Group Name	Configured Frame Size	Applied Frame Size	Frame Rate	Tx port type
1		CE2-R7	Src/Dest Endpoint P...	CE1-R5	EVPL-Vlan-1-EndpointSet-1-Flow Group 0001	Fixed: 64	Fixed: 70	2000 fps	Ethernet
2		CE1-R5	Src/Dest Endpoint P...	CE2-R7	EVPL-Vlan-1-EndpointSet-1-Flow Group 0002	Fixed: 64	Fixed: 70	2000 fps	Ethernet

Traffic Item	Tx Frames	Rx Frames	Frames Delta	Loss %	Tx Frame Rate	Rx Frame Rate	Tx L1 Rate (Bps)	Rx L1 Rate (Bps)	Tx Bytes	Tx Rate (Bps)	Rx Rate (Bps)	Tx Rate (Kbps)	Rx Rate (Kbps)	Tx Rate (Mbps)	Rx Rate (Mbps)
EVPL-Vlan-1	78,149	78,126	23	0.029	4,000.063	3,994.965	2,880,045.594	2,876,086.521	5,468,820	280,004.433	279,619.523	2,240.035...	2,236.956...	2,240.035	2,236.956

1.6 EVPN EVPL Test Scenarios

1.6.1 EVPN EVPL DualTag Configuration – Active/Active mode.

The PE1, PE2 configuration remains the same for most part, just the vlan tags configuration different as highlighted below.

PE1:

```

1 regress@r1-re0# show interfaces ae0
2 ae0 {
3     flexible-vlan-tagging;
4     encapsulation flexible-ethernet-services;
5     esi {
6         00:11:11:11:11:11:11:11:11:11;
7         all-active;
8     }
9     aggregated-ether-options {
10         lacp {
11             active;
12             system-id 00:00:00:00:11:01;
13         }
14     }
15     unit 1 {
16         encapsulation vlan-ccc;
17         vlan-tags outer 1 inner 2;
18     }
19 }

```

PE2:

```
1 regress@r2-re0# show interfaces ae0
2 ae0 {
3     flexible-vlan-tagging;
4     encapsulation flexible-ethernet-services;
5     esi {
6         00:11:11:11:11:11:11:11:11:11;
7         all-active;
8     }
9     aggregated-ether-options {
10         lacp {
11             active;
12             system-id 00:00:00:00:11:01;
13         }
14     }
15     unit 1 {
16         encapsulation vlan-ccc;
17         vlan-tags outer 1 inner 2;
18     }
19 }
```

1.6.2 EVPN EVPL Operational – Active/Active mode

The operational mode for this case is similar to test scenario covered above with single tag.

Traffic Screenshot:

Transmit State	Suspend	Tx Port	Src/Dst Endpoint Pair	Rx Ports	Flow Group Name	Configured Frame Size	Applied Frame Size	Frame Rate	Tx port type
1		R5-CE1	Src/Dst Endpoint P...	R7-CE2	VPWS_EVPL_Dual_Vlan1,2...	Fixed: 64	Fixed: 74	2000 fps	Ethernet
2		R7-CE2	Src/Dst Endpoint P...	R5-CE1	VPWS_EVPL_Dual_Vlan1,2...	Fixed: 64	Fixed: 74	2000 fps	Ethernet

Traffic Item	Tx Frames	Rx Frames	Frames Delta	Loss %	Tx Frame Rate	Rx Frame Rate	Tx L1 Rate (bps)	Rx L1 Rate (bps)	Rx Bytes	Tx Rate (Bps)	Rx Rate (Bps)	Tx Rate (bps)	Rx Rate (bps)	Tx Rate (Kbps)
VPWS_EVPL_Dual_Vlan1,2	121,928	121,909	19	0.016	4,000.183	4,000.683	3,008,137.886	3,008,513.763	9,021,266	296,013.569	296,050.556	2,368,108...	2,368,404...	2,368.109

1.6.3 EVPN EVPL – FAT Flow

The PE1, PE2 configuration remains the same for most part, just the highlighted part in the configuration enables the FAT Flow for the routing instance. Also the Control word is enabled by default and this needs to be disabled if remote PE's do not support control word or it is disabled by default.

PE1:

```
1 regress@r2-RE0-re0# show routing-instances
2 evpn-vpws-epl {
3     instance-type evpn-vpws;
4     protocols {
5         evpn {
6             interface ae0.1 {
7                 vpws-service-id {
8                     local 1;
9                     remote 2;
10                }
11            }
12            flow-label-transmit-static;
13            flow-label-receive-static;
14        }
15    }
16    interface ae0.1;
17    route-distinguisher 22.22.22.22:1;
18    vrf-target target:11:1;
19 }
```


1.7.1 EVPN VPWS 16000 instances Scaling - Script

```

1  # while-else loop
2
3  outervlan = 1
4  ifl = 1
5  local = 1
6  remote = 16001
7  file1 = open('r1.txt', 'w')
8  file2 = open('r2.txt', 'w')
9  file3 = open('r3.txt', 'w')
10 file4 = open('r4.txt', 'w')
11 file5 = open('r5.txt', 'w')
12 file6 = open('r7.txt', 'w')
13 while outervlan < 3:
14     innervlan = 1
15     while innervlan < 4001:
16         #print("set interfaces ae0 unit ", ifl, " vlan-tags outer ", outervlan, " inner ",
17             innervlan)
18         s = "set interfaces ae0 unit " + str(ifl) + " vlan-tags outer " + str(outervlan) + " inner "
19             + str(innervlan) + "\n"
20         s0 = "set interfaces ae0 unit " + str(ifl) + " encapsulation vlan-bridge " + "\n"
21         s1 = "set routing-instances evpn-vpws16k" + str(ifl) + " instance-type evpn-vpws interface "
22             ae0." + str(ifl) + "\n"
23         s3 = "set routing-instances evpn-vpws16k" + str(ifl) + " vrf-target target:11:" + str(ifl)
24             + "\n"
25         s8 = "set interfaces ae0 unit " + str(ifl) + " encapsulation vlan-ccc" + "\n"
26         r1s2 = "set routing-instances evpn-vpws16k" + str(ifl) + " route-distinguisher "
27             11.11.11.11:" + str(ifl) + "\n"
28         r1s4 = "set routing-instances evpn-vpws16k" + str(ifl) + " protocols evpn interface ae0." +
29             str(ifl) + " vpws-service-id local " + str(local) + " remote " + str(remote) + "\n"
30         r2s2 = "set routing-instances evpn-vpws16k" + str(ifl) + " route-distinguisher "
31             22.22.22.22:" + str(ifl) + "\n"
32         r2s4 = "set routing-instances evpn-vpws16k" + str(ifl) + " protocols evpn interface ae0." +
33             str(ifl) + " vpws-service-id local " + str(local) + " remote " + str(remote) + "\n"
34         r3s2 = "set routing-instances evpn-vpws16k" + str(ifl) + " route-distinguisher "
35             33.33.33.33:" + str(ifl) + "\n"
36         r3s4 = "set routing-instances evpn-vpws16k" + str(ifl) + " protocols evpn interface ae0." +
37             str(ifl) + " vpws-service-id local " + str(remote) + " remote " + str(local) + "\n"
38         r4s2 = "set routing-instances evpn-vpws16k" + str(ifl) + " route-distinguisher "
39             44.44.44.44:" + str(ifl) + "\n"
40         r4s4 = "set routing-instances evpn-vpws16k" + str(ifl) + " protocols evpn interface ae0." +
41             str(ifl) + " vpws-service-id local " + str(remote) + " remote " + str(local) + "\n"
42         s5 = "set bridge-domains bd" + str(ifl) + " vlan-tags outer " + str(outervlan) + " inner "
43             + str(innervlan) + "\n"
44         s6 = "set bridge-domains bd" + str(ifl) + " interface ae0." + str(ifl) + "\n"
45         s7 = "set bridge-domains bd" + str(ifl) + " bridge-options no-mac-learning" + "\n"
46         r5s1 = "set interfaces ge-0/0/2 unit " + str(ifl) + " vlan-tags outer " + str(outervlan) +
47             " inner " + str(innervlan) + "\n"
48         r5s2 = "set interfaces ge-0/0/2 unit " + str(ifl) + " encapsulation vlan-bridge " + "\n"
49         r5s3 = "set bridge-domains bd" + str(ifl) + " interface ge-0/0/2." + str(ifl) + "\n"
50         r7s1 = "set interfaces ge-0/0/0 unit " + str(ifl) + " vlan-tags outer " + str(outervlan) +
51             " inner " + str(innervlan) + "\n"
52         r7s2 = "set interfaces ge-0/0/0 unit " + str(ifl) + " encapsulation vlan-bridge " + "\n"
53         r7s3 = "set bridge-domains bd" + str(ifl) + " interface ge-0/0/0." + str(ifl) + "\n"
54
55         file1.write(s)
56         file1.write(s1)
57         file1.write(s3)
58         file1.write(s8)
59         file1.write(r1s2)
60         file1.write(r1s4)
61
62         file2.write(s)
63         file2.write(s1)
64         file2.write(s3)
65         file2.write(s8)
66         file2.write(r2s2)
67         file2.write(r2s4)
68
69         file3.write(s)
70         file3.write(s1)
71         file3.write(s3)
72         file3.write(s8)
73         file3.write(r3s2)
74         file3.write(r3s4)
75
76         file4.write(s)
77         file4.write(s1)

```

```

63     file4.write(s3)
64     file4.write(s8)
65     file4.write(r4s2)
66     file4.write(r4s4)
67
68     file5.write(s)
69     file5.write(s0)
70     file5.write(s5)
71     file5.write(s6)
72     file5.write(s7)
73     file5.write(r5s1)
74     file5.write(r5s2)
75     file5.write(r5s3)
76
77     file6.write(s)
78     file6.write(s0)
79     file6.write(s5)
80     file6.write(s6)
81     file6.write(s7)
82     file6.write(r7s1)
83     file6.write(r7s2)
84     file6.write(r7s3)
85
86     innervlan += 1
87     ifl += 1
88     local += 1
89     remote += 1
90     outervlan += 1

```

1.7.2 EVPN VPWS 500 instances Scaling with Traffic - Operational

PE1

```

1 regress@r2-RE0-re0# run show bgp summary | match "Estab|bgp.evpn"
2 bgp.evpn.0
3 11.11.11.11 64512 13787 4732 0 0 5:42:26 Establ
4   bgp.evpn.0: 1006/1006/1006/0
5 33.33.33.33 64512 1852 2428 0 1 21:49 Establ
6   bgp.evpn.0: 1089/1089/1089/0
7 44.44.44.44 64512 3511 4608 0 0 5:39:12 Establ
8   bgp.evpn.0: 1042/1042/1042/0

```

PE2

```

1 regress@r2-RE0-re0# run show bgp summary | match "Estab|bgp.evpn"
2 bgp.evpn.0
3 11.11.11.11 64512 13787 4732 0 0 5:42:26 Establ
4   bgp.evpn.0: 1006/1006/1006/0
5 33.33.33.33 64512 1852 2428 0 1 21:49 Establ
6   bgp.evpn.0: 1089/1089/1089/0
7 44.44.44.44 64512 3511 4608 0 0 5:39:12 Establ
8   bgp.evpn.0: 1042/1042/1042/0

```

PE3

```

1 regress@r3-RE0-re0# run show bgp summary | match "Estab|bgp.evpn"
2 bgp.evpn.0
3 11.11.11.11 64512 5196 1256 0 0 21:42 Establ
4   bgp.evpn.0: 1005/1005/1005/0
5 22.22.22.22 64512 2429 1850 0 0 21:51 Establ
6   bgp.evpn.0: 1005/1005/1005/0
7 44.44.44.44 64512 1403 1835 0 0 21:48 Establ
8   bgp.evpn.0: 1043/1043/1043/0

```

PE4

```

1 regress@r4-RE0-re0# run show bgp summary | match "Estab|bgp.evpn"
2 bgp.evpn.0
3 11.11.11.11 64512 10828 3126 0 0 5:39:22 Establ
4   bgp.evpn.0: 1005/1005/1005/0
5 22.22.22.22 64512 4609 3509 0 0 5:39:17 Establ
6   bgp.evpn.0: 1005/1005/1005/0
7 33.33.33.33 64512 1837 1402 0 1 21:50 Establ

```

```
8    bgp.evpn.0: 1090/1090/1090/0
```

Traffic snapshot

The screenshot shows the Wireshark interface with the Traffic Item Statistics table selected. The table displays various statistics for the traffic item 'VPWS_EVPL_DualTag_500_services'. The columns include Traffic Item, Tx Frames, Rx Frames, Frames Delta, Loss %, Tx Frame Rate, Rx Frame Rate, Tx L1 Rate (bps), Rx L1 Rate (bps), Tx Bytes, Rx Bytes, Tx Rate (Bps), Rx Rate (Bps), Tx Rate (Mbps), and Rx Rate (Mbps). The data shows a high volume of traffic with a loss percentage of 0.242%.

Traffic Item	Tx Frames	Rx Frames	Frames Delta	Loss %	Tx Frame Rate	Rx Frame Rate	Tx L1 Rate (bps)	Rx L1 Rate (bps)	Tx Bytes	Rx Bytes	Tx Rate (Bps)	Rx Rate (Bps)	Tx Rate (Mbps)	Rx Rate (Mbps)
VPWS_EVPL_DualTag_500_services	63,640	63,486	154	0.242	3,999.533	3,984.035	3,007,649.007	2,995,994.367	4,697,964	295,965.461	294,818.595	2,367,723....	2,358,548....	2,358,548....

1.7.3 EVPN VPWS 16000 instances Scaling - Operational

PF1

```

1 regress@r1-RE0-re0> show bgp summary | match "Establ|bgp.evpn"
2 bgp.evpn.0
3 22.22.22.22 64512 35452 36153 0 0 13:52:00 Establ
4   bgp.evpn.0: 16081/16081/16081/0
5
6 33.33.33.33 64512 17945 68065 0 0 13:52:56 Establ
7   bgp.evpn.0: 16081/16081/16081/0
8 44.44.44.44 64512 34105 45012 0 0 13:51:58 Establ
9   bgp.evpn.0: 16080/16080/16080/0
10
11 regress@r1-RE0-re0> show evpn vpws-instance | match "all-active Primary Up" | count
12 Count: 16000 lines
13
14 regress@r1-RE0-re0> show evpn vpws-instance
15 Instance: evpn-vpws16k1, Instance type: EVPN VPWS
16 Route Distinguisher: 11.11.11.11:1
17 Number of local interfaces: 1 (1 up)
18
19   Interface name ESI Mode Role Status Control-Word Flow-Label-Tx Flow-Label-Rx
20   ae0.1 00:11:11:11:11:11:11:11:11:11:11:11:11:11:11:11 all-active Primary Up Yes No No
21     Local SID: 1 Advertised Label: 47
22       PE addr ESI Label Mode Role TS Status
23       22.22.22.22 00:11:11:11:11:11:11:11:11:11:11 53 all-active Primary 2022-12-05 23:35:56.191
24         Resolved
25     Remote SID: 16001
26       PE addr ESI Label Mode Role TS Status
27       44.44.44.44 00:22:22:22:22:22:22:22:22:22:22 47 all-active Primary 2022-12-05 23:35:56.636
28         Resolved
29       33.33.33.33 00:22:22:22:22:22:22:22:22:22:22 51 all-active Primary 2022-12-06 00:16:34.623
30         Resolved
31 Number of protect interfaces: 0
32
33   Fast Convergence Information
34   ESI: 00:11:11:11:11:11:11:11:11:11:11:11:11:11:11:11 Number of PE nodes: 1
35     PE: 22.22.22.22
36     Advertised SID: 1
37
38   Fast Convergence Information
39   ESI: 00:22:22:22:22:22:22:22:22:22:22:22:22:22:22:22 Number of PE nodes: 2
40     PE: 44.44.44.44
41     Advertised SID: 16001
42     PE: 33.33.33.33
43     Advertised SID: 16001

```

PE2

```
1 regress@r2-RE0-re0> show bgp summary | match "Establ|bgp.evpn"
2 bgp.evpn.0
3 11.11.11.11 64512 36168 35464 0 0 13:58:14 Establ
4     bgp.evpn.0: 16081/16081/16081/0
5 33.33.33.33 64512 17949 70060 0 0 13:58:07 Establ
6     bgp.evpn.0: 16081/16081/16081/0
7 44.44.44.44 64512 34116 52969 0 0 13:58:11 Establ
```

```

8      bgp.evpn.0: 16080/16080/16080/0
9
10     regress@r2-RE0-re0> show evpn vpws-instance
11 Instance: evpn-vpws16k1, Instance type: EVPN VPWS
12 Route Distinguisher: 22.22.22.22:1
13 Number of local interfaces: 1 (1 up)
14   Interface name ESI Mode Role Status Control-Word Flow-Label-Tx Flow-Label-Rx
15 ae0.1 00:11:11:11:11:11:11:11:11:11:11:11:11:11:11 all-active Primary Up Yes No No
16   Local SID: 1 Advertised Label: 53
17     PE addr ESI Label Mode Role TS Status
18   11.11.11.11 00:11:11:11:11:11:11:11:11:11:11:11:11:11 47 all-active Primary 2022-12-05 23:34:28.430
19     Resolved
20   Remote SID: 16001
21     PE addr ESI Label Mode Role TS Status
22   44.44.44.44 00:22:22:22:22:22:22:22:22:22:22:22:22:22 47 all-active Primary 2022-12-05 23:36:08.229
23     Resolved
24   33.33.33.33 00:22:22:22:22:22:22:22:22:22:22:22:22:22 51 all-active Primary 2022-12-06 00:16:34.613
25     Resolved
26 Number of protect interfaces: 0
27
28 Fast Convergence Information
29 ESI: 00:11:11:11:11:11:11:11:11:11:11:11:11:11 Number of PE nodes: 1
30 PE: 11.11.11.11
31   Advertised SID: 1
32
33 Fast Convergence Information
34 ESI: 00:22:22:22:22:22:22:22:22:22:22:22:22:22 Number of PE nodes: 2
35 PE: 44.44.44.44
36   Advertised SID: 16001
37 PE: 33.33.33.33
38   Advertised SID: 16001
39
40 regress@r2-RE0-re0> show evpn vpws-instance | match "all-active Primary Up" | count
41 Count: 16000 lines

```

PE3

```

1 regress@r3-RE0-re0> show bgp summary | match "Establ|bgp.evpn"
2 bgp.evpn.0
3 11.11.11.11 64512 68086 17963 0 0 14:01:33 Establ
4   bgp.evpn.0: 16080/16080/16080/0
5 22.22.22.22 64512 70067 17953 0 0 14:00:30 Establ
6   bgp.evpn.0: 16080/16080/16080/0
7 44.44.44.44 64512 67805 17957 0 0 14:01:25 Establ
8   bgp.evpn.0: 16081/16081/16081/0
9
10 {master}
11 regress@r3-RE0-re0> show evpn vpws-instance
12 Instance: evpn-vpws16k1, Instance type: EVPN VPWS
13 Route Distinguisher: 33.33.33.33:1
14 Number of local interfaces: 1 (1 up)
15
16 Interface name ESI Mode Role Status Control-Word Flow-Label-Tx Flow-Label-Rx
17 ae0.1 00:22:22:22:22:22:22:22:22:22:22:22 all-active Primary Up Yes No No
18 Local SID: 16001 Advertised Label: 51
19 PE addr ESI Label Mode Role TS Status
20 44.44.44.44 00:22:22:22:22:22:22:22:22:22:22:22 47 all-active Primary 2022-12-06 00:09:55.581
    Resolved
21 Remote SID: 1
22 PE addr ESI Label Mode Role TS Status
23 22.22.22.22 00:11:11:11:11:11:11:11:11:11:11:11 53 all-active Primary 2022-12-06 00:09:39.917
    Resolved
24 11.11.11.11 00:11:11:11:11:11:11:11:11:11:11:11 47 all-active Primary 2022-12-06 00:09:46.478
    Resolved
25 Number of protect interfaces: 0
26
27 Fast Convergence Information
28 ESI: 00:11:11:11:11:11:11:11:11:11:11:11 Number of PE nodes: 2
29 PE: 22.22.22.22
30   Advertised SID: 1
31 PE: 11.11.11.11
32   Advertised SID: 1
33
34 Fast Convergence Information
35 ESI: 00:22:22:22:22:22:22:22:22:22:22:22 Number of PE nodes: 1
36 PE: 44.44.44.44
37   Advertised SID: 16001
38
39 {master}
40 regress@r3-RE0-re0> show evpn vpws-instance | match "all-active Primary Up" | count

```

```
41 Count: 16000 lines
```

PE4

```
1 regress@r4-RE0-re0> show bgp summary | match "Establ|bgp.evpn"
2 bgp.evpn.0
3 11.11.11.11 64512 45099 34188 0 0 14:30:45 Establ
4 bgp.evpn.0: 16080/16080/16080/0
5 22.22.22.22 64512 53042 34186 0 0 14:30:43 Establ
6 bgp.evpn.0: 16080/16080/16080/0
7 33.33.33.33 64512 18024 67869 0 0 14:31:35 Establ
8 bgp.evpn.0: 16082/16082/16082/0
9
10 {master}
11 regress@r4-RE0-re0> show evpn vpws-instance
12 Instance: evpn-vpws16k1, Instance type: EVPN VPWS
13 Route Distinguisher: 44.44.44.44:1
14 Number of local interfaces: 1 (1 up)
15
16 Interface name ESI Mode Role Status Control-Word Flow-Label-Tx Flow-Label-Rx
17 ae0.1 00:22:22:22:22:22:22:22:22all-active Primary Up Yes No No
18 Local SID: 16001 Advertised Label: 47
19 PE addr ESI Label Mode Role TS Status
20 33.33.33.33 00:22:22:22:22:22:22:22:22 51 all-active Primary 2022-12-06 00:16:34.651
21 Resolved
22 Remote SID: 1
23 PE addr ESI Label Mode Role TS Status
24 11.11.11.11 00:11:11:11:11:11:11:11:11 47 all-active Primary 2022-12-05 23:34:28.424
25 Resolved
26 22.22.22.22 00:11:11:11:11:11:11:11:11 53 all-active Primary 2022-12-05 23:34:33.433
27 Resolved
28 Number of protect interfaces: 0
29
30 Fast Convergence Information
31 ESI: 00:11:11:11:11:11:11:11:11 Number of PE nodes: 2
32 PE: 11.11.11.11
33 Advertised SID: 1
34 PE: 22.22.22.22
35 Advertised SID: 1
36
37 Fast Convergence Information
38 ESI: 00:22:22:22:22:22:22:22:22 Number of PE nodes: 1
39 PE: 33.33.33.33
40 Advertised SID: 16001
41 {master}
42 regress@r4-RE0-re0> show evpn vpws-instance | match "all-active Primary Up" | count
43 Count: 16000 lines
```

1.8 Configuration Snippets

Underlay Configuration – MPLS/OSPF

PE1:

```
1 regress@r1-re0# show interfaces
2 et-0/0/1 {
3     unit 0 {
4         description to-r4;
5         family inet {
6             address 14.1.1.1/30;
7         }
8         family mpls;
9     }
10 }
11 et-0/0/3 {
12     unit 0 {
13         description to-r2;
14         family inet {
15             address 12.1.1.1/30;
16         }
17         family mpls;
18     }
19 }
20 et-0/0/5 {
```

```

21     unit 0 {
22         description to-r3;
23         family inet {
24             address 13.1.1.1/30;
25         }
26         family mpls;
27     }
28 }
29 et-0/0/7 {
30     description "To R5";
31     ether-options {
32         802.3ad ae0;
33     }
34 }
35 lo0 {
36     unit 0 {
37         family inet {
38             address 11.11.11.11/32;
39         }
40     }
41 }
42
43 regress@r1-re0# show protocols
44 bgp {
45     group ibgp {
46         type internal;
47         local-address 11.11.11.11;
48         family evpn {
49             signaling;
50         }
51         neighbor 44.44.44.44;
52         neighbor 33.33.33.33;
53         neighbor 22.22.22.22;
54     }
55 }
56 ldp {
57     interface all;
58     interface fxp0.0 {
59         disable;
60     }
61 }
62 mpls {
63     interface all;
64     interface fxp0.0 {
65         disable;
66     }
67 }
68 ospf {
69     area 0.0.0.0 {
70         interface et-0/0/1.0;
71         interface et-0/0/3.0;
72         interface et-0/0/5.0;
73         interface lo0.0 {
74             passive;
75         }
76     }
77 }
78 lldp {
79     interface all;
80 }
81
82 regress@r1-re0# show routing-options
83 router-id 11.11.11.11;
84 autonomous-system 64512;

```

PE2:

```

1 regress@r2-re0# show interfaces
2 et-0/0/1 {
3     unit 0 {
4         description to-r3;
5         family inet {
6             address 23.1.1.1/30;
7         }
8         family mpls;
9     }
10 }
11 et-0/0/3 {
12     unit 0 {
13         description to-r1;

```

```

14     family inet {
15         address 12.1.1.2/30;
16     }
17     family mpls;
18 }
19 }
20 et-0/0/5 {
21     unit 0 {
22         description to-r4;
23         family inet {
24             address 24.1.1.1/30;
25         }
26         family mpls;
27     }
28 }
29 et-0/0/9 {
30     description "To R5";
31     ether-options {
32         802.3ad ae0;
33     }
34 }
35 lo0 {
36     unit 0 {
37         family inet {
38             address 22.22.22.22/32;
39         }
40     }
41 }
42
43 regress@r2-re0# show protocols
44 bgp {
45     group ibgp {
46         type internal;
47         local-address 22.22.22.22;
48         family evpn {
49             signaling;
50         }
51         neighbor 11.11.11.11;
52         neighbor 44.44.44.44;
53         neighbor 33.33.33.33;
54     }
55 }
56 ldp {
57     interface all;
58     interface fxp0.0 {
59         disable;
60     }
61 }
62 mpls {
63     interface all;
64     interface fxp0.0 {
65         disable;
66     }
67 }
68 ospf {
69     area 0.0.0.0 {
70         interface et-0/0/1.0;
71         interface et-0/0/3.0;
72         interface et-0/0/5.0;
73         interface lo0.0 {
74             passive;
75         }
76     }
77 }
78 lldp {
79     interface all;
80 }
81
82 regress@r2-re0# show routing-options
83 router-id 22.22.22.22;
84 autonomous-system 64512;

```

PE3:

```

1 regress@r3-re0# show interfaces
2 et-0/0/1 {
3     unit 0 {
4         description to-r2;
5         family inet {
6             address 23.1.1.2/30;

```

```

7      }
8      family mpls;
9  }
10 }
11 et-0/0/3 {
12     unit 0 {
13         description to-r1;
14         family inet {
15             address 13.1.1.2/30;
16         }
17         family mpls;
18     }
19 }
20 et-0/0/5 {
21     unit 0 {
22         description to-r4;
23         family inet {
24             address 34.1.1.1/30;
25         }
26         family mpls;
27     }
28 }
29 et-0/0/7 {
30     description "To R7";
31     ether-options {
32         802.3ad ae0;
33     }
34 }
35 lo0 {
36     unit 0 {
37         family inet {
38             address 33.33.33.33/32;
39         }
40     }
41 }
42
43 regress@r3-re0# show protocols
44 bgp {
45     group ibgp {
46         type internal;
47         local-address 33.33.33.33;
48         family evpn {
49             signaling;
50         }
51         neighbor 11.11.11.11;
52         neighbor 22.22.22.22;
53         neighbor 44.44.44.44;
54     }
55 }
56 ldp {
57     interface all;
58     interface fxp0.0 {
59         disable;
60     }
61 }
62 mpls {
63     interface all;
64     interface fxp0.0 {
65         disable;
66     }
67 }
68 ospf {
69     area 0.0.0.0 {
70         interface et-0/0/1.0;
71         interface et-0/0/3.0;
72         interface et-0/0/5.0;
73         interface lo0.0 {
74             passive;
75         }
76     }
77 }
78 lldp {
79     interface all;
80 }
81
82 regress@r3-re0# show routing-options
83 router-id 33.33.33.33;
84 autonomous-system 64512;

```

PE4:

```

1 regress@r4-re0# show interfaces
2 et-0/0/1 {
3     unit 0 {
4         description to-r1;
5         family inet {
6             address 14.1.1.2/30;
7         }
8         family mpls;
9     }
10 }
11 et-0/0/3 {
12     unit 0 {
13         description to-r2;
14         family inet {
15             address 24.1.1.2/30;
16         }
17         family mpls;
18     }
19 }
20 et-0/0/5 {
21     unit 0 {
22         description to-r3;
23         family inet {
24             address 34.1.1.2/30;
25         }
26         family mpls;
27     }
28 }
29 et-0/0/7 {
30     description "To R7";
31     ether-options {
32         802.3ad ae0;
33     }
34 }
35 lo0 {
36     unit 0 {
37         family inet {
38             address 44.44.44.44/32;
39         }
40     }
41 }
42
43 regress@r4-re0# show protocols
44 bgp {
45     group ibgp {
46         type internal;
47         local-address 44.44.44.44;
48         family evpn {
49             signaling;
50         }
51         neighbor 11.11.11.11;
52         neighbor 22.22.22.22;
53         neighbor 33.33.33.33;
54     }
55 }
56 ldp {
57     interface all;
58     interface fxp0.0 {
59         disable;
60     }
61 }
62 mpls {
63     interface all;
64     interface fxp0.0 {
65         disable;
66     }
67 }
68 ospf {
69     area 0.0.0.0 {
70         interface et-0/0/1.0;
71         interface et-0/0/3.0;
72         interface et-0/0/5.0;
73         interface lo0.0 {
74             passive;
75         }
76     }
77 }
78 lldp {
79     interface all;
80 }

```

```

81
82 regress@r4-re0# show routing-options
83 router-id 44.44.44.44;
84 autonomous-system 64512;

```

Underlay Configuration – SR, ISIS

PE1:

```

1 regress@r1-re0# show interfaces
2 et-0/0/1 {
3     unit 0 {
4         description to-r4;
5         family inet {
6             address 14.1.1.1/30;
7         }
8         family mpls;
9         family iso;
10    }
11 }
12 et-0/0/3 {
13     unit 0 {
14         description to-r2;
15         family inet {
16             address 12.1.1.1/30;
17         }
18         family mpls;
19         family iso;
20    }
21 }
22 et-0/0/5 {
23     unit 0 {
24         description to-r3;
25         family inet {
26             address 13.1.1.1/30;
27         }
28         family mpls;
29         family iso;
30    }
31 }
32 et-0/0/7 {
33     description "To R5";
34     ether-options {
35         802.3ad ae0;
36     }
37 }
38 lo0 {
39     unit 0 {
40         family inet {
41             address 11.11.11.11/32;
42         }
43         family iso {
44             address 49.0001.1111.1111.00
45         }
46     }
47 }
48
49 regress@r1-re0# show protocols
50 bgp {
51     group ibgp {
52         type internal;
53         local-address 11.11.11.11;
54         family evpn {
55             signaling;
56         }
57         neighbor 44.44.44.44;
58         neighbor 33.33.33.33;
59         neighbor 22.22.22.22;
60     }
61 }
62 isis {
63     interface all {
64         point-to-point;
65     }
66     interface fxp0.0 {
67         disable;
68     }
69     source-packet-routing {
70         node-segment ipv4-index 1001;

```

```

71     }
72     level 2 wide-metrics-only;
73     level 1 disable;
74 }
75 mpls {
76     interface all;
77     interface fxp0.0 {
78         disable;
79     }
80 }
81 lldp {
82     interface all;
83 }
84
85 regress@r1-re0# show routing-options
86 router-id 11.11.11.11;
87 autonomous-system 64512;

```

PE2:

```

1 regress@r2-re0# show interfaces
2 et-0/0/1 {
3     unit 0 {
4         description to-r3;
5         family inet {
6             address 23.1.1.1/30;
7         }
8         family mpls;
9         family iso;
10    }
11 }
12 et-0/0/3 {
13     unit 0 {
14         description to-r1;
15         family inet {
16             address 12.1.1.2/30;
17         }
18         family mpls;
19         family iso;
20    }
21 }
22 et-0/0/5 {
23     unit 0 {
24         description to-r4;
25         family inet {
26             address 24.1.1.1/30;
27         }
28         family mpls;
29         family iso;
30    }
31 }
32 et-0/0/9 {
33     description "To R5";
34     ether-options {
35         802.3ad ae0;
36     }
37 }
38 lo0 {
39     unit 0 {
40         family inet {
41             address 22.22.22.22/32;
42         }
43         family iso {
44             address 49.0001.2222.2222.00
45         }
46     }
47 }
48
49 regress@r2-re0# show protocols
50 bgp {
51     group ibgp {
52         type internal;
53         local-address 22.22.22.22;
54         family evpn {
55             signaling;
56         }
57         neighbor 11.11.11.11;
58         neighbor 44.44.44.44;
59         neighbor 33.33.33.33;
60     }

```

```

61 }
62 isis {
63     interface all {
64         point-to-point;
65     }
66     interface fxp0.0 {
67         disable;
68     }
69     source-packet-routing {
70         node-segment ipv4-index 1002;
71     }
72     level 2 wide-metrics-only;
73     level 1 disable;
74 }
75 mpls {
76     interface all;
77     interface fxp0.0 {
78         disable;
79     }
80 }
81 lldp {
82     interface all;
83 }
84
85 regress@r2-re0# show routing-options
86 router-id 22.22.22.22;
87 autonomous-system 64512;

```

PE3:

```

1 regress@r3-re0# show interfaces
2 et-0/0/1 {
3     unit 0 {
4         description to-r2;
5         family inet {
6             address 23.1.1.2/30;
7         }
8         family mpls;
9         family iso;
10    }
11 }
12 et-0/0/3 {
13     unit 0 {
14         description to-r1;
15         family inet {
16             address 13.1.1.2/30;
17         }
18         family mpls;
19         family iso;
20    }
21 }
22 et-0/0/5 {
23     unit 0 {
24         description to-r4;
25         family inet {
26             address 34.1.1.1/30;
27         }
28         family mpls;
29         family iso;
30    }
31 }
32 et-0/0/7 {
33     description "To R7";
34     ether-options {
35         802.3ad ae0;
36     }
37 }
38 lo0 {
39     unit 0 {
40         family inet {
41             address 33.33.33.33/32;
42         }
43         family iso {
44             address 49.0001.3333.3333.00
45         }
46     }
47 }
48
49 regress@r3-re0# show protocols
50 bgp {

```

```

51     group ibgp {
52         type internal;
53         local-address 33.33.33.33;
54         family evpn {
55             signaling;
56         }
57         neighbor 11.11.11.11;
58         neighbor 22.22.22.22;
59         neighbor 44.44.44.44;
60     }
61 }
62 isis {
63     interface all {
64         point-to-point;
65     }
66     interface fxp0.0 {
67         disable;
68     }
69     source-packet-routing {
70         node-segment ipv4-index 1003;
71     }
72     level 2 wide-metrics-only;
73     level 1 disable;
74 }
75
76 mpls {
77     interface all;
78     interface fxp0.0 {
79         disable;
80     }
81 }
82 lldp {
83     interface all;
84 }
85
86 regress@r3-re0# show routing-options
87 router-id 33.33.33.33;
88 autonomous-system 64512;

```

PE4:

```

1 regress@r4-re0# show interfaces
2 et-0/0/1 {
3     unit 0 {
4         description to-r1;
5         family inet {
6             address 14.1.1.2/30;
7         }
8         family mpls;
9         family iso;
10    }
11 }
12 et-0/0/3 {
13     unit 0 {
14         description to-r2;
15         family inet {
16             address 24.1.1.2/30;
17         }
18         family mpls;
19         family iso;
20    }
21 }
22 et-0/0/5 {
23     unit 0 {
24         description to-r3;
25         family inet {
26             address 34.1.1.2/30;
27         }
28         family mpls;
29         family iso;
30    }
31 }
32 et-0/0/7 {
33     description "To R7";
34     ether-options {
35         802.3ad ae0;
36     }
37 }
38 lo0 {
39     unit 0 {

```

```

40     family inet {
41         address 44.44.44.44/32;
42     }
43     family iso {
44         address 49.0001.4444.4444.00
45     }
46 }
47 }
48
49 regress@r4-re0# show protocols
50 bgp {
51     group ibgp {
52         type internal;
53         local-address 44.44.44.44;
54         family evpn {
55             signaling;
56         }
57         neighbor 11.11.11.11;
58         neighbor 22.22.22.22;
59         neighbor 33.33.33.33;
60     }
61 }
62 isis {
63     interface all {
64         point-to-point;
65     }
66     interface fxp0.0 {
67         disable;
68     }
69     source-packet-routing {
70         node-segment ipv4-index 1004;
71     }
72     level 2 wide-metrics-only;
73     level 1 disable;
74 }
75 mpls {
76     interface all;
77     interface fxp0.0 {
78         disable;
79     }
80 }
81 lldp {
82     interface all;
83 }
84
85 regress@r4-re0# show routing-options
86 router-id 44.44.44.44;
87 autonomous-system 64512;

```

CE Configuration in EPL, A/A mode

CE1:

```

1 regress@r5_re0# show interfaces
2 ge-0/0/0 {
3     description "To R2";
4     gigether-options {
5         802.3ad ae0;
6     }
7 }
8 ge-0/0/1 {
9     description "To R1";
10    gigether-options {
11        802.3ad ae0;
12    }
13 }
14 ge-0/0/2 {
15     description "To IXIA - (10.49.119.55)";
16     flexible-vlan-tagging;
17     encapsulation flexible-ethernet-services;
18     unit 0 {
19         encapsulation ethernet-bridge;
20     }
21 }
22 ae0 {
23     description "Connected to PE1";
24     flexible-vlan-tagging;
25     encapsulation flexible-ethernet-services;
26     aggregated-ether-options {

```

```

27     lacp {
28         active;
29         system-id 00:00:00:00:11:02;
30     }
31 }
32 unit 1 {
33     encapsulation ethernet-bridge;
34 }
35 }

```

CE2:

```

1 regress@r7_re0# show interfaces
2 ge-0/0/0 {
3     description "To IXIA --";
4     flexible-vlan-tagging;
5     encapsulation flexible-ethernet-services;
6     unit 0 {
7         encapsulation ethernet-bridge;
8     }
9 }
10 ge-0/0/1 {
11     description "To R3";
12     gigether-options {
13         802.3ad ae0;
14     }
15 }
16 ge-0/0/2 {
17     description "To R4";
18     gigether-options {
19         802.3ad ae0;
20     }
21 }
22 ae0 {
23     description "Connected to PE3-4";
24     flexible-vlan-tagging;
25     encapsulation flexible-ethernet-services;
26     aggregated-ether-options {
27         lacp {
28             active;
29         }
30     }
31     unit 0 {
32         encapsulation ethernet-bridge;
33     }
34 }

```

CE Configuration in EPL, A/S mode

CE1:

```

1 regress@r5_re0# show interfaces
2 ge-0/0/0 {
3     description "To R2";
4     encapsulation ethernet-bridge;
5     unit 0 {
6         family bridge;
7     }
8 }
9 ge-0/0/1 {
10    description "To R1";
11    encapsulation ethernet-bridge;
12    unit 0 {
13        family bridge;
14    }
15 }
16 ge-0/0/2 {
17    description "To IXIA - (10.49.119.55)";
18    encapsulation ethernet-bridge;
19    unit 0 {
20        family bridge;
21    }
22 }
23 regress@r5_re0# show bridge-domains
24 bd1 {
25     interface ge-0/0/2.0;
26     interface ge-0/0/0.0;
27     interface ge-0/0/1.0;

```

```

28     bridge-options {
29         no-mac-learning;
30     }
31 }

```

CE2:

```

1 regress@r7_re0# show interfaces
2 ge-0/0/0 {
3     description "To IXIA-R10 - 10.49.106.252 ";
4     inactive: flexible-vlan-tagging;
5     encapsulation ethernet-bridge;
6     unit 0 {
7         family bridge;
8     }
9 }
10 ge-0/0/1 {
11     description "To R3";
12     encapsulation ethernet-bridge;
13     unit 0 {
14         family bridge;
15     }
16 }
17 ge-0/0/2 {
18     description "To R4";
19     encapsulation ethernet-bridge;
20     unit 0 {
21         family bridge;
22     }
23 }
24 regress@r7_re0# show bridge-domains
25 bd1 {
26     interface ge-0/0/0.0;
27     interface ge-0/0/1.0;
28     interface ge-0/0/2.0;
29     bridge-options {
30         no-mac-learning;
31     }
32 }

```

CE Configuration in EPL & Dual-Tag with A/A mode

CE1:

```

1 regress@R5-RE0> show configuration interfaces
2 ge-0/0/0 {
3     description "To R2";
4     gigether-options {
5         802.3ad ae0;
6     }
7 }
8 ge-0/0/1 {
9     description "To R1";
10    gigether-options {
11        802.3ad ae0;
12    }
13 }
14 ge-0/0/2 {
15     description "To IXIA";
16     flexible-vlan-tagging;
17     encapsulation flexible-ethernet-services;
18     unit 1 {
19         encapsulation vlan-bridge;
20         vlan-tags outer 1 inner 2;
21     }
22 }
23 ae0 {
24     description "Connected to PE1";
25     flexible-vlan-tagging;
26     encapsulation flexible-ethernet-services;
27     aggregated-ether-options {
28         lacp {
29             active;
30             system-id 00:00:00:00:11:02;
31         }
32     }
33     unit 1 {
34         encapsulation vlan-bridge;

```

```

35     vlan-tags outer 1 inner 2;
36 }
37 }
38
39 regress@R5-RE0> show configuration bridge-domains
40 bdl {
41     vlan-tags outer 1 inner 2;
42     interface ae0.1;
43     interface ge-0/0/2.1;
44     bridge-options {
45         no-mac-learning;
46     }
47 }

```

CE2:

```

1 regress@R7-RE0> show configuration interfaces
2 ge-0/0/0 {
3     description "To IXXIA";
4     flexible-vlan-tagging;
5     encapsulation flexible-ethernet-services;
6     unit 1 {
7         encapsulation vlan-bridge;
8         vlan-tags outer 1 inner 2;
9     }
10 }
11 ge-0/0/1 {
12     description "To R3";
13     gigether-options {
14         802.3ad ae0;
15     }
16 }
17 ge-0/0/2 {
18     description "To R4";
19     gigether-options {
20         802.3ad ae0;
21     }
22 }
23 ae0 {
24     description "Connected to PE1";
25     flexible-vlan-tagging;
26     encapsulation flexible-ethernet-services;
27     aggregated-ether-options {
28         lacp {
29             active;
30             system-id 00:00:00:00:22:02;
31         }
32     }
33     unit 1 {
34         encapsulation vlan-bridge;
35         vlan-tags outer 1 inner 2;
36     }
37 }
38
39 regress@R7-RE0> show configuration bridge-domains
40 bdl {
41     vlan-tags outer 1 inner 2;
42     interface ae0.1;
43     interface ge-0/0/0.1;
44     bridge-options {
45         no-mac-learning;
46     }
47 }

```

1.9 Glossary

- CE: Customer Edge
- DF: Designated Forwarder
- ESI: Ethernet Segment Identifier
- EVI: EVPN Instance
- EVPN: Ethernet VPN
- MOD: Modular Function

- NDF: Non Designated Forwarder
- PE: Provider Edge
- VPWS: Virtual Private Wired Service

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